Higher-Order Statistics in Simulation-Based Inference for Non-Gaussian Stochastic Gravitational-Wave Backgrounds

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[Abstract to be rewritten once the positioning under referee comment 3 is fixed.] Standard searches for the stochastic gravitational-wave background (SGWB) rely on cross-correlation statistics that optimally extract signal power under the assumption that the background is Gaussian and stationary. While valid for the superposition of many overlapping sources, this assumption breaks down for intermittent populations, such as high-redshift compact binary coalescences or cosmic-string cusps in the low-duty-cycle regime. We present a simulation-based inference framework, ING-Net (Intermittent Non-Gaussian Network), that addresses this regime by training a neural posterior estimator on a compact summary vector,

$$x = (C_{\rm HL},\, P,\, \kappa_{\rm H1},\, \kappa_{\rm L1})\,,$$

in which the per-detector excess kurtosis terms capture fourth-cumulant information beyond what is accessible to second-order power-spectrum-based summaries. We validate the pipeline on real detector noise from the LIGO O3a, O3b, and O4a observing runs, with injections that explicitly model the antenna-pattern modulation of the H1–L1 detector pair. For low-duty-cycle signals, $\xi \ll 1$, ING-Net alleviates the rate–amplitude degeneracy that limits Gaussian-spectral methods and reaches a sensitivity in $\Omega_{\rm GW}$ that is approximately 0.66 orders of magnitude deeper than a matched-architecture scalar baseline at $\xi = 10^{-3}$ across all three runs. A controlled comparison against a frequency-resolved baseline of comparable input dimensionality shows that this gain is not a feature-count effect but reflects the information content of the fourth cumulant; the per-detector form of the kurtosis additionally implements an implicit coherency veto against single-detector glitches. On noise-only data, the non-Gaussian framework yields upper limits on $\Omega_{\rm GW}$ within $|\Delta_{\rm paired}| \leq 0.09\,$dex of the spectral baseline, confirming that the additional statistics do not introduce false signals.

I. INTRODUCTION

The detection of the Stochastic Gravitational-Wave Background (SGWB) represents the next major frontier in gravitational-wave astronomy. Unlike the resolvable "chirp" signals from individual compact binary mergers detected by the LIGO-Virgo-KAGRA (LVK) collaboration [1, 2], the SGWB is formed by the incoherent superposition of unresolved sources. These sources are expected to span a wide range of physical origins, from the astrophysical population of binary black holes (BBH) and neutron stars (BNS) to cosmological processes such as inflation, first-order phase transitions, and cosmic strings [3–5].

To date, searches for the SGWB have primarily relied on the optimal cross-correlation statistic, often referred to as the Y-statistic [6, 7]. This method assumes that the background is isotropic, stationary, and Gaussian. The Gaussian assumption is rigorously justified by the Central Limit Theorem (CLT) when the number of overlapping sources is large. Under these conditions, the background is fully characterized by its second-order moment, the Power Spectral Density (PSD) or equivalently the dimensionless energy density spectrum $\Omega_{\rm GW}(f)$.

However, the Gaussian assumption is not universally valid. Several theoretically motivated sources produce backgrounds that are highly intermittent, often referred to as "popcorn" noise or shot noise. For example, the population of massive black hole binaries at high redshift ($z > 2$) may have merger rates low enough that signals do not significantly overlap in the detector frame [4, 8]. Similarly, gravitational waves from cusps and kinks on cosmic string loops are expected to arrive as distinct, short-duration single events [9, 10]. In this low duty-cycle regime ($\xi \ll 1$), the signal is strongly non-Gaussian.

Traditional spectral methods, which average energy over long observation times ($T_{\rm obs}$), effectively dilute these sparse high-energy bursts into the noise floor. While optimal for Gaussian signals, the cross-correlation statistic is sub-optimal for intermittent signals because it discards the phase information and higher-order moments that characterize the temporal morphology of the bursts [9, 11].

Addressing this limitation requires a method capable of extracting information from the temporal structure of the signal. While analytic likelihoods for non-Gaussian noise can be derived for specific simplified models (e.g., Drasco & Flanagan [9]), they are often computationally intractable for realistic data or difficult to adapt to complex, non-stationary detector noise artifacts (glitches) which also exhibit non-Gaussian behavior.

In this paper, we present a robust, data-driven approach using Simulation-Based Inference (SBI) [12]. SBI, also known as Likelihood-Free Inference, utilizes neural networks to approximate the posterior distribution directly from forward simulations, bypassing the need for an explicit likelihood function [13, 14]. By training a Normalizing Flow [15] on summary statistics that include the kurtosis of the strain data, our framework learns to identify the statistical signature of intermittency. We demonstrate that this method can distinguish between a weak continuous background and a strong intermittent one—a distinction not accessible to second-order spectral summaries alone—and provides superior sensitivity

using real LIGO data.

II. METHODOLOGY

A. Signal Model: The Transition Regime

We model the isotropic SGWB strain signal $h(t)$ in a detector as a superposition of independent transient events. Mathematically, this can be expressed as:

$$h(t) = \sum_{i=1}^{N} A_i w(t - t_i), \tag{1}$$

where t_i are the arrival times governed by a Poisson process with rate R, A_i are the amplitudes drawn from a source-dependent distribution, and $w(t)$ is the normalized waveform of a single event.

We instantiate this model with a deliberately generic event. Each event is a band-limited Gaussian burst of fixed duration $\tau = 8/f_s \simeq 3.9$ ms ($f_s = 2048$ Hz), with a Hann envelope $w(t)$ normalised so that $\langle w^2 \rangle = 1$ over the burst, and with flat power in the whitened domain between $f_{\rm lo} = 20$ Hz and $f_{\rm hi} = 1000$ Hz. Event start times form a Poisson process of rate R; events may straddle segment boundaries, so that the expected fraction of time covered by signal is exactly the duty cycle

$$\xi = R\,\tau, \tag{2}$$

and the expected number of events in a segment of length T is

$$\langle N_{\rm burst} \rangle = R\,T = \xi\,T/\tau, \tag{3}$$

i.e. $\langle N_{\rm burst} \rangle \simeq 1.0$ per $T = 4$ s segment at $\xi = 10^{-3}$. Overlapping events add incoherently. The per-event variance in the whitened, unit-variance domain is $(\Omega/\xi)\,\sigma_1^2$, where σ_1^2 is the per-sample variance produced by a Gaussian background of $\Omega_{\rm GW} = 1$ [Eq. (11)]; the time-averaged energy density is therefore Ω, independent of ξ. The burst carries no source-specific waveform or spectral colouring, so that sensitivity to intermittency is driven by how the energy is partitioned in time rather than by the morphology of individual events; the assumptions and limitations of this choice are discussed in Sec. VI B.

The injection accounts for the antenna response of the H1–L1 pair. For every event we draw an independent sky direction (isotropic) and polarisation angle $\psi \in [0, \pi)$, compute the H1 and L1 antenna factors $F^I_{+,\times}$ from the detector tensors [16], and inject $h^I = F^I_+ h_+ + F^I_\times h_\times$ with independent Gaussian polarisation realisations. A single constant rescales the injected power so that Ω retains its meaning as the total isotropic energy density; the zero-lag H1–L1 overlap reduction function of the injected signal is $\gamma_{\rm HL}(0) \simeq -0.89$, in agreement with the standard value for the H1–L1 pair [17]. We also report results for an "identical" injection, in which the same realisation is added to both detectors, to separate the effect of the antenna response (Sec. VI B).

The statistical morphology of this background is primarily controlled by the duty cycle ξ of Eq. (2). This parameter defines a continuous spectrum of signal types. In the *Gaussian regime* ($\xi \gg 1$), the arrival rate is high and events overlap significantly, so the resulting distribution of $h(t)$ converges to a Gaussian via the Central Limit Theorem—the standard "confusion noise" background. In the *intermittent regime* ($\xi \ll 1$), events are sparse and non-overlapping, producing a probability density function with heavy tails (leptokurtic distribution). This is the "popcorn" regime.

Figure 2 illustrates this transition by contrasting two regimes side-by-side: (a) a Gaussian-like SGWB, where the strain time series is featureless and the amplitude distribution is Gaussian; and (b) an intermittent SGWB, where the duty cycle ξ is small enough that individual bursts are visible against the noise floor and the amplitude distribution develops heavy non-Gaussian tails. Standard cross-correlation pipelines, which rely solely on second-moment information, cannot distinguish (a) from (b) because their second moments are matched; per-detector kurtosis statistics, by contrast, separate the two, since only the intermittent regime produces elevated kurtosis in *both* H1 and L1. The same per-detector kurtosis pair additionally discriminates a genuine astrophysical signal from a single-detector instrumental glitch—a transient present in the real detector noise rather than an injected signal class—which elevates the kurtosis of only one channel. This is the central motivation for the feature set we introduce in Sec. II C; the way the per-detector kurtosis pair additionally separates genuine signals from single-detector glitches is examined in Sec. VI A.

B. Statistical Theory of Non-Gaussianity

To understand why standard methods fail, it is useful to consider the Edgeworth expansion of the probability density function $p(x)$ for the strain data. For a zero-mean process, the PDF can be approximated as:

$$p(x) \approx G(x)\left[1 + \frac{\kappa_3}{6} H_3(x) + \frac{\kappa_4}{24} H_4(x) + \dots\right], \tag{4}$$

where $G(x)$ is the Gaussian distribution, $H_n(x)$ are the Hermite polynomials, and κ_n are the standardized cumulants.

Standard cross-correlation methods implicitly maximize a Gaussian likelihood, which corresponds to keeping only the first term in the expansion (where $\kappa_n = 0$ for $n > 2$). For intermittent signals, however, the fourth cumulant is non-zero and grows as the duty cycle decreases. For the model of Sec. II, in the limit of non-overlapping events ($\xi \ll 1$) the signal at a given sample is non-zero with probability ξ, and its local variance is $(\Omega/\xi)\sigma_1^2\,A\,w^2$, where A is the (normalised) antenna power of the event.

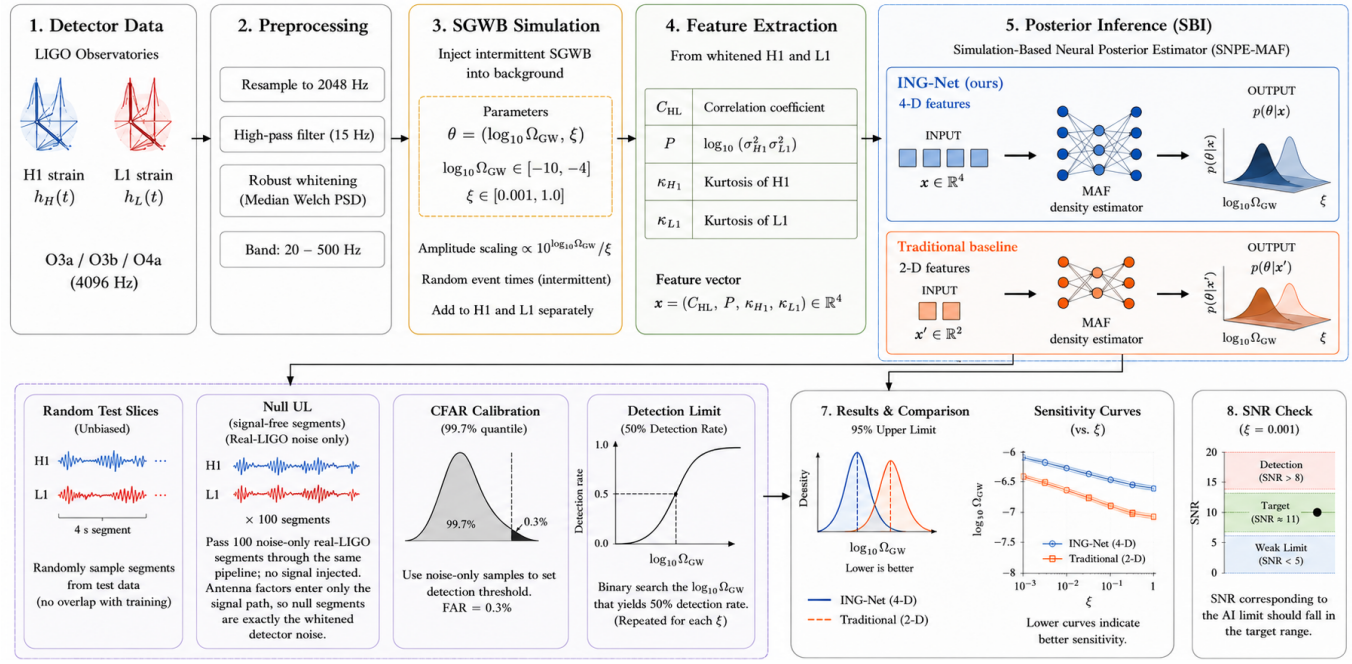


FIG. 1: End-to-end pipeline for ING-Net, organised into four stages (numbered panels). (1) Raw H1 and L1 strain from the O3a, O3b, and O4a observing runs. (2) Data conditioning: resampling to 2048 Hz, high-pass filtering at 15 Hz, and median-Welch whitening per time block, band-limited to 20–1000 Hz, with robust unit-variance normalisation, producing 4 s segments of whitened detector noise. (3) Antenna-pattern signal injection (Sec. II): plus and cross polarisations $h_+(t), h_\times(t)$ are drawn from an intermittent-SGWB simulator with prior $\log_{10}\Omega \in [-8, -1]$, $\log_{10}\xi \in [-3, 0]$; an independent sky direction (θ, ϕ) and polarisation angle ψ are drawn isotropically per event; and the per-detector antenna factors $F^I_{+,\times}(\theta, \phi, \psi)$ project the polarisations onto each detector, with the resulting strain added to the corresponding whitened noise segment. This reproduces the standard H1–L1 overlap reduction function $\gamma_{\rm HL}(0) \approx -0.89$ and a mean squared antenna response $\langle F^2\rangle \approx 0.20$. (4) The signal-injected segments are compressed into a 4-D feature vector $x = (C_{HL}, P, \kappa_{H1}, \kappa_{L1})$ (ING-Net) or a 2-D vector $x' = (C_{HL}, P)$ (power-only baseline), and each is passed through a Masked Autoregressive Flow with identical architecture; only the input dimensionality differs, isolating the information content of the kurtosis statistics. Detection thresholds are calibrated on noise-only segments drawn from the time-disjoint evaluation part of each stretch (Sec. II C) at the 99.7% CFAR level, and sensitivity limits are obtained by binary search for the 50%-detection amplitude $\Omega_{\rm limit}$.

The excess kurtosis of the signal alone is then

$$\kappa_{\rm sig} = 3\left(\frac{c_w\, c_F}{\xi} - 1\right), \tag{5}$$

where $c_w = \langle w^4\rangle/\langle w^2\rangle^2 \simeq 1.73$ is set by the burst envelope and $c_F = \langle A^2\rangle/\langle A\rangle^2 \simeq 1.43$ by the isotropic antenna response, with $c_F = 1$ for identical injection. What is measured, however, is the kurtosis of signal plus noise. For a signal of time-averaged per-sample variance $\sigma_s^2 = \Omega\sigma_1^2$ added to unit-variance noise of excess kurtosis κ_n,

$$\kappa_{\rm obs} = \frac{\kappa_{\rm sig}\,\sigma_s^4 + \kappa_n}{(\sigma_s^2 + 1)^2}, \tag{6}$$

so near the detection limit ($\sigma_s^2 \ll 1$) the signal contribution is suppressed by σ_s^4 and must compete with both the estimator variance, $\mathrm{Var}(\hat\kappa) \simeq 24/N$ for N samples, and the intrinsic non-Gaussianity of the detector noise. Equation (6) shows why the kurtosis channel is informative only at small ξ, where $\kappa_{\rm sig} \propto 1/\xi$ can compensate the σ_s^4 suppression.

C. Data Conditioning and Feature Extraction

We analyse 15 stretches of LIGO H1 and L1 strain of 4096 s each, five from each of O3a, O3b and O4a: one reference stretch per run (Table I) and four further stretches per run drawn at random by a rule fixed before any of them was analysed (Appendix A: data-quality flags set in both detectors, no published candidate event in the stretch, and a minimum separation of one day; no stretch was discarded after selection). Each stretch is resampled from 4096 Hz to $f_s = 2048$ Hz and high-pass filtered at 15 Hz, and then split in time: the first 75% is used to generate training simulations and the last 25%, separated by an 8 s gap, is used for everything downstream of training (CFAR calibration, sensitivity injections and

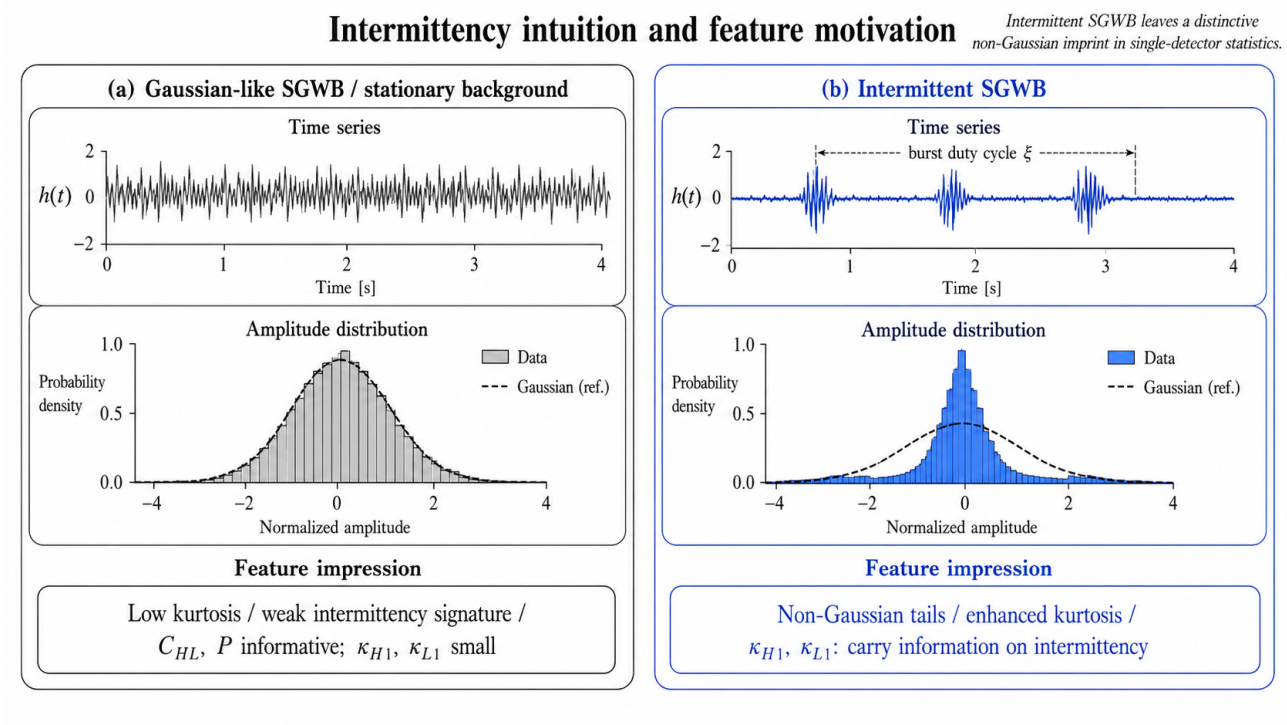


FIG. 2: Intermittency intuition and feature motivation. Two contrasting regimes are shown in side-by-side panels, each with the strain time series (top), normalised amplitude distribution (middle), and the qualitative feature impression that the per-detector summary statistics carry (bottom). *(a) Gaussian-like SGWB:* stationary background, amplitude histogram tracks a Gaussian, kurtosis is low; the cross-correlation C_{HL} and log-power P carry the relevant information and the kurtosis channels κ_{H1}, κ_{L1} are uninformative. *(b) Intermittent SGWB:* the duty cycle ξ is small and individual bursts are visible in the time series, producing pronounced non-Gaussian tails and large kurtosis in *both* detectors. The discriminating information for the intermittent regime lives in the kurtosis channel and is invisible to spectral methods; the complementary role of the per-detector kurtosis in rejecting single-detector glitches is discussed in Sec. VI A.

posterior checks). Each part is whitened separately with its own median-averaged Welch PSD, restricted to $f_{\rm lo}$–$f_{\rm hi}$ = 20–1000 Hz, trimmed by 16 s at both ends to remove filter transients, and scaled to unit variance using a robust (median-absolute-deviation) scale estimate, so that a small number of loud transients does not set the normalisation. This leaves 3040 s of training noise and 984 s of evaluation noise per stretch, with no sample shared between them. The whitened streams contain real instrumental noise, including glitches, and no coherent astrophysical signal; the intermittent background is injected as described in Sec. II.

To perform inference, we compress the high-dimensional time-series data d into a four-dimensional vector of summary statistics $x(d)$, motivated by the theoretical work of Brillinger [18] on polyspectra. The first component is the cross-correlation coefficient between the two detectors,

$$C_{HL} = \frac{\langle d_H d_L \rangle}{\sqrt{\langle d_H^2 \rangle \langle d_L^2 \rangle}}, \tag{7}$$

which leverages the coherence of the SGWB across the baseline to reject incoherent detector noise. The second is the log-power P, proportional to the variance of the data and the standard statistic for constraining the amplitude $\Omega_{\rm ref}$. The third and fourth components are the excess kurtosis of the H1 and L1 channels individually, defined as

$$\kappa = \frac{E[(d-\mu)^4]}{(E[(d-\mu)^2])^2} - 3. \tag{8}$$

For a Gaussian distribution, $\kappa = 0$; for intermittent signals, $\kappa \gg 0$. The inclusion of per-detector kurtosis is the critical innovation that enables our network to constrain ξ: it provides sensitivity to the temporal clustering of signal energy that spectral methods discard.

We use *per-detector* kurtosis κ_{H1}, κ_{L1} rather than a cross-detector cumulant such as the cross-kurtosis or the trispectrum. Keeping the two channels separate preserves the information needed to tell whether excess kurtosis appears in both detectors at once or in only one, so the four-dimensional feature vector $(C_{HL}, P, \kappa_{H1}, \kappa_{L1})$ carries strictly more information than either a Gaussian-stochastic spectral analysis or a single-channel non-Gaussianity test. A useful by-product of this choice—an implicit coherency veto against single-detector glitches

that emerges without any hand-designed veto criterion—is discussed in Sec. VI A.

D. Role of Summary Statistics and Information Content

A central design choice in our framework is the selection of summary statistics. Our pipeline compresses each 4-second data segment into a four-dimensional vector: the cross-correlation coefficient C_{HL}, the log-power P, and the excess kurtosis of the H1 and L1 channels (κ_{H1}, κ_{L1}). This choice is motivated by the Edgeworth expansion (Eq. 3), which shows that the leading non-Gaussian correction enters at the level of the fourth cumulant.

Our power-only baseline conditions the same neural posterior estimator on the two summary statistics that carry the information used by a standard cross-correlation analysis—the inter-detector correlation C_{HL} and the log-power P—while ING-Net additionally receives the per-detector kurtosis pair. The two pipelines share the MAF architecture, hyperparameters, and training procedure and differ *only* in input dimensionality, so the performance gap in Table II isolates the information content of the kurtosis statistics rather than any architectural difference. Section VI A extends this comparison to a frequency-resolved baseline whose input dimensionality matches or exceeds ING-Net's; it reaches the same sensitivity as the scalar baseline and does not close the gap to ING-Net at low ξ.

An important subtlety in our implementation is the use of *individual-detector* kurtosis values (κ_{H1} and κ_{L1}) rather than their cross-correlation. This design choice enables the network to implement an implicit "coherency veto": an astrophysical non-Gaussian signal produces correlated excess kurtosis in both detectors, whereas an instrumental glitch (which typically affects only one detector) produces excess kurtosis in a single channel. The network can learn to distinguish these two cases by comparing κ_{H1} and κ_{L1}, effectively rejecting instrumental contamination without explicit glitch subtraction.

Nevertheless, our choice of summary statistics is not exhaustive. The third cumulant (skewness) could provide additional constraining power for asymmetric burst amplitude distributions. The bispectrum and trispectrum, which are the frequency-domain analogs of the third and fourth cumulants [18], carry phase information that our time-domain kurtosis discards. Wavelet-based features could capture the time-frequency morphology of individual bursts. Whether these additional statistics would improve sensitivity, or merely enlarge the feature space without adding information, remains an open question.

III. SIMULATION-BASED INFERENCE

A. Neural Posterior Estimation (NPE)

We employ Neural Posterior Estimation (NPE) to approximate the posterior distribution $p(\theta|x)$, where $\theta = [\log_{10}\Omega, \log_{10}\xi]$ are the physical parameters.

NPE is naturally suited to the kind of problem posed by intermittent SGWBs. The Gaussian-mixture likelihood of [9], in which each per-segment data vector is modelled as a draw from one of two Gaussian components (signal-present or signal-absent), gives a closed-form likelihood only after one commits to specific per-component covariances; once the burst morphology, antenna modulation, or non-stationary detector noise is included exactly, the likelihood becomes intractable. NPE sidesteps this by learning a flexible posterior estimator from forward simulations, requiring only that the simulator produce statistically representative data. The role of the kurtosis summary in our framework is essentially to give the density estimator low-dimensional access to the very excess fourth cumulant that the two-component mixture of [9] is built to produce, while letting the network handle the realistic noise and antenna physics that the closed-form likelihood cannot.

The goal of NPE is to learn a density estimator $q_\phi(\theta|x)$, parameterized by weights ϕ, that approximates the true posterior. The network is trained by minimizing the Kullback–Leibler (KL) divergence between the true posterior and the approximation, which is equivalent to maximizing the log-probability of the true parameters under the network's output:

$$\mathcal{L}(\phi) = -\mathbb{E}_{(\theta,x)\sim p(\theta,x)}[\log q_\phi(\theta|x)] \tag{9}$$

We note one consequence of this design choice. NPE directly approximates the posterior $p(\theta|x)$ rather than the model evidence $Z = \int p(x|\theta)\, p(\theta)\, d\theta$, so the Bayes factor between two SGWB models is not immediately available from the trained density estimator and would require a separate evidence-estimation step (e.g. importance sampling against the learned posterior). For the present work we therefore quantify the relative performance of the ING-Net and Trad pipelines through the sensitivity-limit gap Δ in $\log_{10}\Omega_{\rm limit}$ rather than through a Bayes-factor comparison. The two metrics carry related information—a deeper achievable upper limit at fixed false-alarm rate corresponds to a model that more decisively separates the signal and noise hypotheses—but Δ has the operational advantage of mapping directly onto the physical question ("what amplitude of Ω can be excluded") and is well defined for both density-estimating (NPE) and frequentist (cross-correlation) pipelines, enabling the like-for-like comparison reported in Table II.

B. Network Architecture

We utilize a Masked Autoregressive Flow (MAF) [19] as the density estimator, implemented via the `sbi` toolkit [13]. The MAF is constructed from a stack of 5 MADE (Masked Autoencoder for Distribution Estimation) blocks, each consisting of a 2-layer MLP with 50 hidden units per layer and ReLU activation functions.

The training set consists of 5×10^4 forward simulations drawn from a prior uniform in $\log_{10}\Omega \in [-8, -1]$ and in $\log_{10}\xi \in [-3, 0]$, so that each decade of duty cycle receives equal training mass. The noise realisations are drawn from the training part of each stretch only (Sec. II C), and the signal is injected through the H1–L1 antenna response as described in Sec. II. The network was trained using the Adam optimizer with a learning rate of 5×10^{-4}, a batch size of 256, and early stopping with a patience of 30 epochs.

Training-seed variability. MAF training on GPU is not bit-deterministic: with the default PyTorch settings (`cudnn.deterministic` not enforced), reduction-order variability in the underlying CUBLAS kernels causes two runs with identical seed, data, and hyperparameters to produce numerically different network weights. We characterise the magnitude of this effect empirically by training each network configuration with three independent seeds (1400–1402) on every stretch. For ING-Net (4-D summary) the per-cell seed-to-seed scatter is $\sigma_{\text{seed}} \lesssim [X]$ dex; for the scalar Trad baseline (2-D summary) it is $\sigma_{\text{seed}} \lesssim [Y]$ dex. [Headline numbers: definition of the average over seeds and stretches to be fixed with Table II.]

IV. VALIDATION ON MOCK DATA

Before applying our pipeline to the real LIGO strain of Sec. V, we performed a controlled sanity check on synthetic Gaussian colored noise. This stage uses a simpler, fully controlled noise environment to verify that the posterior is statistically well-calibrated and that the kurtosis-conditioned network recovers the duty cycle under the intermittent-signal model considered here; the real-detector-noise results follow in Sec. V.

A. Posterior Calibration

To ensure the reliability of our Bayesian inference, we performed a coverage test using Probability-Probability (P-P) plots. For a well-calibrated posterior, the true parameter value should fall within the $p\%$ credible interval exactly $p\%$ of the time.

Figure 3(a) presents the P-P plot for the duty cycle parameter ξ. The curve closely follows the diagonal $y = x$, indicating that the posterior probabilities are well-calibrated. The network is neither over-confident nor under-confident in its uncertainties.

TABLE I: Dataset configurations and training parameters. The reference stretch of each run is listed by its GPS start time; the remaining stretches are listed in Appendix A.

	O3a	O3b	O4a
Reference GPS start	1238919012	1264612209	1369205931
Stretches	5	5	5
Stretch duration (s)	4096	4096	4096
Train / eval. noise (s)	3040 / 984 per stretch		
Sample rate (Hz)	2048	2048	2048
Training seeds	1400–1402 per stretch		
Training samples	5×10^4 per model		
$\log_{10}\Omega$, $\log_{10}\xi$ prior	$\mathcal{U}[-8,-1]$, $\mathcal{U}[-3,0]$		

B. Duty-Cycle Recovery

Figure 3(b) compares the duty-cycle estimates of the kurtosis-conditioned neural posterior estimator—ING-Net—with those of a "power-only" baseline conditioned only on the second-order summaries. ING-Net (blue dots) recovers the true duty cycle across the parameter space, whereas the estimates of the power-only baseline (orange squares) are uncorrelated with the truth.

This outcome is expected by construction. Because the per-event variance scales as Ω/ξ, the time-averaged power is independent of ξ, so the second-order summaries (C_{HL}, P) carry essentially no information on the duty cycle and the power-only posterior on ξ must revert to the prior. The comparison in Fig. 3(b) is therefore not evidence for a deficiency of second-order statistics beyond this construction; its role is to show that ING-Net *does* recover ξ from the fourth-cumulant channel. The non-trivial result of this work is the gain in Ω_{limit} (Sec. V).

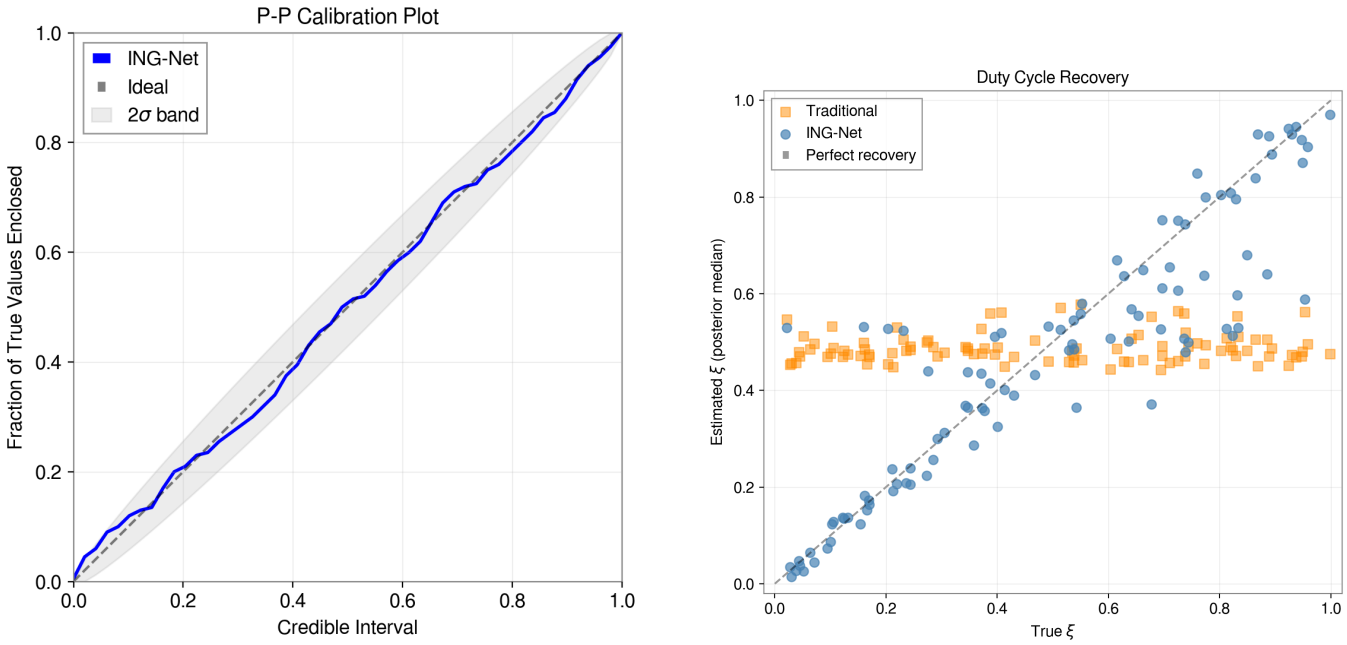
Figure 4 further highlights this for a specific injection of $\xi = 0.011$. ING-Net produces a sharp posterior peak at the true value, effectively pinpointing the intermittency. The power-only baseline returns a broad, flat posterior, reflecting its inability to distinguish a rare, strong signal from a frequent, weak one.

V. RESULTS: APPLICATION TO LIGO DATA

We applied our pipeline to 15 stretches of real data, five from each of the O3a, O3b and O4a observing runs, selected and partitioned in time as described in Sec. II C and Appendix A. The signal is injected through the H1–L1 antenna response (Sec. II). Table I summarizes the datasets and training configurations.

A. Parameter Estimation on Intermittent Signals

We performed injection campaigns with a simulated intermittent signal ($\xi = 0.05$) added to the real noise of



(a) P-P Plot

(b) Truth-vs-Predicted

FIG. 3: Validation metrics. *Left:* P-P plot for the duty cycle parameter, confirming that the ING-Net posterior is well calibrated (consistent with the diagonal within the 2σ band). *Right:* Duty cycle recovery, showing that ING-Net (blue) accurately tracks the true ξ across the full prior range, while the power-only baseline (orange) collapses to a constant $\xi \approx 0.5$ regardless of the true value.

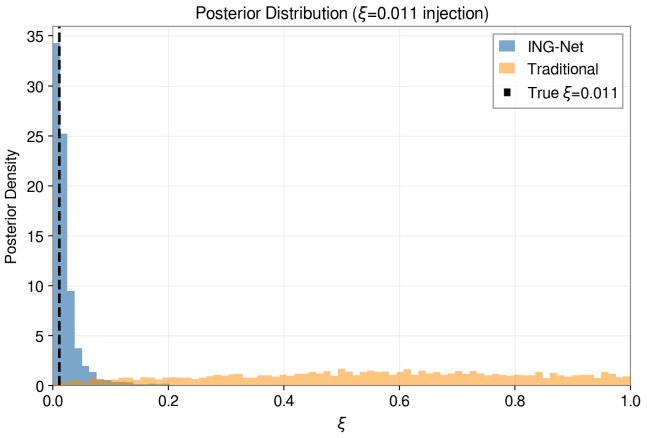


FIG. 4: Posterior distribution for the duty cycle with an injected signal at $\xi = 0.011$. ING-Net (blue) produces a sharp posterior concentrated near the true value, while the power-only baseline (orange) yields a nearly uniform distribution, demonstrating its inability to constrain ξ.

each dataset. Figure 5 displays the joint posterior distribution for the O3a dataset as a representative example.

The ING-Net framework (blue contours) tightly constrains both the amplitude Ω and the duty cycle ξ, with the posterior concentrated near the injected values. The network successfully distinguishes the coherent astrophysical bursts from the incoherent instrumental glitches by exploiting the per-detector kurtosis pair $(\kappa_{H1}, \kappa_{L1})$ as discussed in Sec. II C and Sec. VI A: a glitch typically appears in only one detector and elevates only one of the two channels, whereas a GW burst appears in both and elevates both. Conversely, the power-only baseline (orange contours) suffers from the rate–amplitude degeneracy, with its ξ marginal spanning essentially the entire prior range. This behavior is consistent across all three datasets, and is the visual statement of the rate–amplitude degeneracy that we discuss in Sec. V E.

B. Sensitivity Across Observing Runs

To quantify the performance gain, we computed the sensitivity limit $\Omega_{\rm limit}$ for each dataset using a binary search procedure with CFAR-calibrated thresholds at the 99.7% quantile, as illustrated in Fig. 6. The sensitivity limit is defined as the minimum signal amplitude required for confident detection at 50% probability; each stretch is analysed with three independent training seeds (1400–1402). Table II presents the full results.

The detection statistic is the posterior mean of $\log_{10}\Omega$

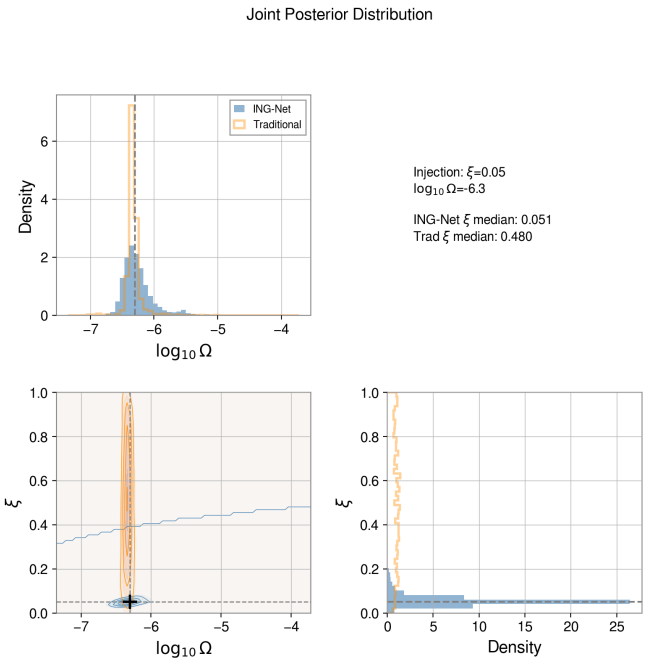


FIG. 5: Joint posterior distribution for an injected signal with $\xi = 0.05$ in O3a noise. ING-Net (blue) tightly constrains both $\log_{10}\Omega$ and ξ near the true values, whereas the power-only baseline (orange) correctly recovers the amplitude but exhibits a broad, uninformative marginal in ξ.

estimated from $n = 200$ samples of the trained flow,

$$s(x) = \frac{1}{n}\sum_{j=1}^{n} \log_{10}\Omega^{(j)}, \qquad (\Omega^{(j)}, \xi^{(j)}) \sim q_\phi(\theta \mid x), \tag{10}$$

evaluated identically for all pipelines. The threshold η_{CFAR} is the 99.7% quantile of $s(x)$ over 10^4 noise-only segments drawn at random start times from the evaluation part of each stretch. Because the evaluation part spans 984 s, these draws overlap and correspond to only $N_{\mathrm{eff}} \approx 246$ independent 4 s segments, so that the 99.7% quantile is effectively set by the loudest one or two independent segments. Treating the threshold as an order statistic of N_{eff} independent draws, the false-alarm probability per segment of the threshold actually used lies in $[0.02\%, 1.2\%]$ (90% interval, median 0.28%) rather than being exactly 0.3%. To check that our conclusions do not hinge on these extreme segments, we repeated the limit calculation with the threshold set at the 99% quantile (≈ 2.5 independent exceedances). The gain Δ changes by -0.01 ± 0.04 dex at $\xi = 10^{-2}$ and -0.04 ± 0.09 dex at $\xi = 10^{-3}$ (paired over 41 runs), and ING-Net remains ahead in all 15 stretches at both duty cycles.

The results reveal two distinct regimes that are consistent across all three observing runs. In the *Gaussian convergence* limit ($\xi \to 1$) the signal becomes effectively continuous and the kurtosis carries no extra information, so the two methods reach comparable sensitivity ($|\Delta| \leq 0.039$). In the *non-Gaussian* regime ($\xi \ll 1$) ING-Net significantly outperforms the baseline across all datasets: at $\xi = 10^{-3}$ the gain is $\Delta = 0.640$–0.683 dex (mean 0.664 dex over the three runs), corresponding to a factor of ~ 4.6 improvement in the detectable signal energy density. [Numbers in this paragraph to be updated with Table II.]

Differences between pipelines are interpreted only when they exceed twice their combined seed-to-seed standard error; smaller offsets, including the sign of Δ in the Gaussian limit, are consistent with zero.

Figure 7 shows the sensitivity curves for the O3a dataset as a representative example. The non-Gaussian advantage is consistently present in the low-ξ regime across all three datasets.

C. Calibration and single-event signal-to-noise ratio

The amplitude axis is calibrated per stretch. In the whitened, band-limited, unit-variance domain, a Gaussian isotropic background with one-sided strain spectrum $S_{\mathrm{GW}}(f;\Omega_{\mathrm{GW}})$ produces a per-sample variance $\Omega_{\mathrm{GW}}\,\sigma_1^2$, with

$$\sigma_1^2 = \frac{1}{f_{\mathrm{hi}} - f_{\mathrm{lo}}} \int_{f_{\mathrm{lo}}}^{f_{\mathrm{hi}}} \frac{S_{\mathrm{GW}}(f;\Omega_{\mathrm{GW}}{=}1)}{P_n(f)}\, df, \tag{11}$$

where P_n is the measured one-sided noise PSD. Both factors in the integrand are spectral densities, so σ_1^2 is dimensionless: it is a ratio of per-sample variances, not an optimal-filter SNR.

The corresponding per-sample amplitude of a single event, $\rho_{\mathrm{ev}} = \sqrt{\Omega/\xi}\,\sigma_1$, should likewise not be read as a signal-to-noise ratio. The optimal (matched-filter) SNR accumulated over one burst of τf_s samples is

$$\rho_{\mathrm{burst}} = \rho_{\mathrm{ev}}\sqrt{\tau f_s\,\langle w^2\rangle}\,\sqrt{A} = \sqrt{\frac{\Omega}{\xi}\,\tau f_s\,A}\;\sigma_1, \tag{12}$$

per detector, where A is the event's normalised antenna power ($\langle A\rangle = 1$). At the detection limits of Table II the median single-event SNR is $\rho_{\mathrm{burst}} \approx 15$ (range 12–22 over runs) at $\xi = 10^{-3}$, where a segment contains on average one event, and $\rho_{\mathrm{burst}} \approx 3.9$ (3.5–6.0) at $\xi = 10^{-2}$, where it contains on average ten. We return to the consequences for the relation with single-event burst searches in Sec. VI C.

D. Upper Limits on Null Results

[Section pending: delete, or replace by a direct count of false alarms on the evaluation noise.] Finally, we analyzed segments of pure noise (the null hypothesis) for each dataset. Table III compares the 95% upper limits

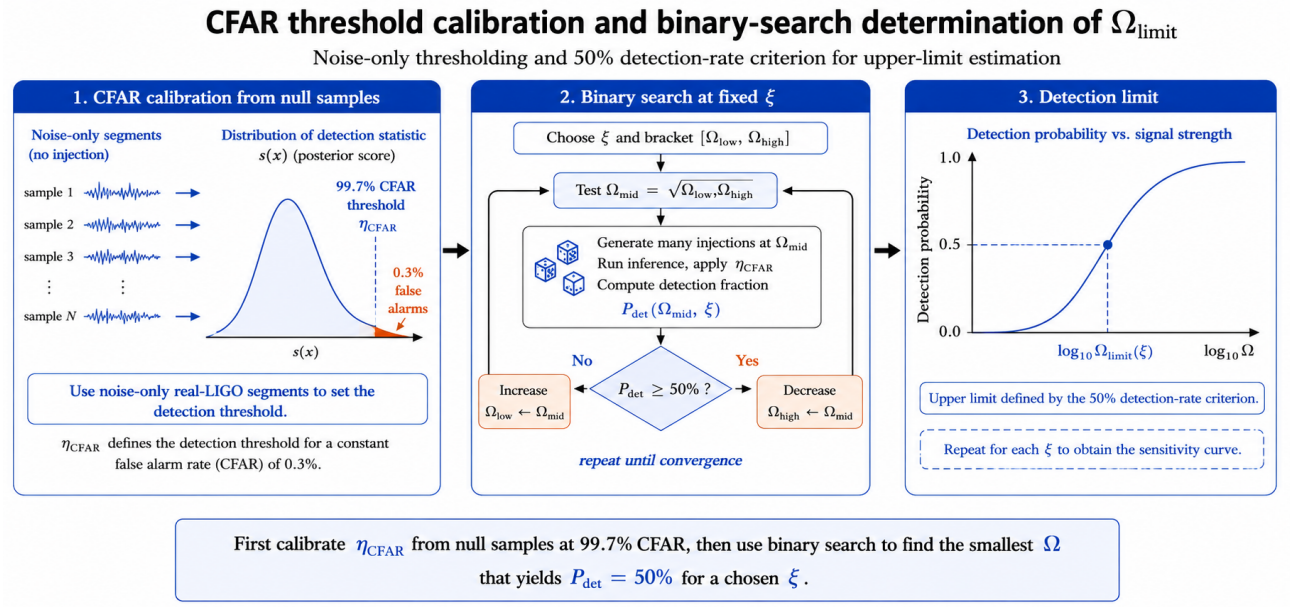


FIG. 6: Threshold calibration and sensitivity-limit estimation procedure. *Panel 1: CFAR calibration.* A large pool of noise-only segments is passed through the trained posterior estimator to produce the distribution of the detection statistic $s(x)$ [Eq. (10)] under the null hypothesis; the 99.7% quantile of this distribution defines the threshold $\eta_{\rm CFAR}$, which corresponds to a nominal 0.3% false-alarm rate. *Panel 2: Binary search at fixed ξ.* For each duty cycle, we bracket the amplitude axis between $\Omega_{\rm low}$ and $\Omega_{\rm high}$, draw many injections at the geometric midpoint $\Omega_{\rm mid} = \sqrt{\Omega_{\rm low}\Omega_{\rm high}}$, score each through the network at the threshold $\eta_{\rm CFAR}$, and compute the detection fraction $P_{\rm det}(\Omega_{\rm mid}, \xi)$. The bracket is halved depending on whether $P_{\rm det}$ exceeds or falls short of 50%, and the procedure iterates until convergence. *Panel 3: Detection limit.* The converged value defines $\log_{10}\Omega_{\rm limit}(\xi)$ as the signal strength yielding $P_{\rm det} = 50\%$. Repeating across ξ produces the sensitivity curve reported in Table II.

on $\log_{10}\Omega$ obtained from ING-Net and the power-only baseline.

Across all three runs the paired offset $|\Delta_{\rm paired}| \leq 0.09$ dex, confirming that ING-Net does not introduce false signals on noise-only data: in the absence of a non-Gaussian component the kurtosis channel carries no information and the posterior collapses to a power-only constraint, recovering the power-only baseline's behaviour.

E. Connecting back to the rate–amplitude degeneracy

Figure 5 illustrates the same construction on real noise: the power-only posterior is constrained in $\log_{10}\Omega$ but follows the line of constant time-averaged power in ξ, whereas ING-Net localises ξ through the fourth cumulant. The degeneracy of second-order statistics—along ξ at fixed Ω in our parametrisation, in which Ω is the time-averaged energy density, or equivalently along $\xi\,\Omega_{\rm ev} = $ const in terms of the per-event energy density $\Omega_{\rm ev} = \Omega/\xi$—is built into the signal model, and has been derived analytically in the Gaussian-mixture framework by Liu and Kuroyanagi [20]; we therefore present the ξ recovery as a property of the estimator rather than as an independent empirical finding.

VI. DISCUSSION

A. Comparison with previous methods and the information content of the kurtosis channel

Previous attempts to detect non-Gaussian backgrounds have largely relied on constructing specialized statistics for particular signal models. Drasco and Flanagan [9] first formulated the Gaussian-mixture likelihood for intermittent SGWB detection, treating each data segment as a mixture of signal-present and signal-absent Gaussian components and deriving a corresponding likelihood-ratio statistic. While theoretically elegant, the analytic tractability of this framework relies on simplifying assumptions about the per-segment signal distribution, and extending it to realistic non-stationary detector noise or to physically motivated burst morphologies typically requires additional model-specific structure. Martellini and Regimbau [11] proposed a semi-parametric framework that avoids explicit waveform templates but requires numerically evaluating higher-order spectral integrals, which becomes computationally expensive for long observation times. More recently,

TABLE II: [Table values and caption to be replaced with the 15-stretch results once the primary configuration is fixed.] Sensitivity limits ($\log_{10}\Omega_{\rm limit}$) for ING-Net and the power-only baseline across three observing runs, with realistic antenna-pattern modulation applied to the H1–L1 detector pair (Sec. II, Sec. VI B). Values are mean ± standard deviation over 5 independent training seeds (1400–1404). $\Delta \equiv \log_{10}\Omega_{\rm limit}^{\rm Trad} - \log_{10}\Omega_{\rm limit}^{\rm ING\text{-}Net}$ denotes the ING-Net advantage in orders of magnitude; positive Δ means ING-Net reaches a deeper limit. For O3b, one of the five Trad seeds (1400) failed to converge—the binary search hit the prior boundary $\log_{10}\Omega = -4$ at every ξ—and is excluded from the Trad column for that run; the ING-Net column uses all five seeds. Excluding a diverged Trad seed is conservative with respect to Δ, since a diverged baseline would inflate it.

Dataset	ξ	ING-Net Mean	ING-Net Std	Trad Mean	Trad Std	Δ
O3a	0.001	−7.340	0.057	−6.657	0.015	+0.683
	0.01	−7.555	0.040	−7.244	0.017	+0.311
	0.1	−7.351	0.027	−7.286	0.015	+0.064
	0.5	−7.318	0.044	−7.296	0.015	+0.022
	1.0	−7.317	0.050	−7.302	0.022	+0.015
O3b	0.001	−7.420	0.068	−6.752	0.029	+0.669
	0.01	−7.619	0.041	−7.325	0.020	+0.295
	0.1	−7.421	0.026	−7.363	0.022	+0.058
	0.5	−7.386	0.035	−7.374	0.012	+0.012
	1.0	−7.381	0.034	−7.379	0.017	+0.002
O4a	0.001	−7.383	0.024	−6.743	0.045	+0.640
	0.01	−7.594	0.030	−7.338	0.014	+0.256
	0.1	−7.377	0.036	−7.359	0.023	+0.018
	0.5	−7.352	0.041	−7.375	0.031	−0.023
	1.0	−7.344	0.030	−7.383	0.016	−0.039

TABLE III: 95% upper limits on $\log_{10}\Omega$ from null (noise-only) analyses, computed over 100 independent noise segments per dataset. $\Delta_{\rm paired}$, reported in the last column, is the mean of per-segment paired differences (Trad − ING-Net); this paired statistic removes segment-level noise variability and is therefore a more sensitive estimator of the systematic offset between methods than a difference of unpaired per-method means.

Dataset	ING-Net mean	ING-Net std	Trad mean	Trad std	$\Delta_{\rm paired}$
O3a	−7.42	0.19	−7.47	0.30	−0.048
O3b	−7.39	0.20	−7.35	0.22	+0.047
O4a	−7.46	0.23	−7.37	0.34	+0.089

Thrane [21] developed a frequentist statistic based on the excess variance of cross-correlation measurements across time segments, demonstrating sensitivity to non-Gaussianity in simulated data. However, this approach does not naturally produce posterior distributions for the physical parameters of the background.

Our SBI framework offers a fundamentally different paradigm. Rather than designing a hand-crafted statistic for a specific signal class, the neural network learns the mapping from summary statistics to the posterior distribution directly from forward simulations. This confers three practical advantages. First, the method is flexible: adapting it to a different signal model requires only modifying the simulator, not re-deriving the optimal statistic. Second, because the simulations are built on real detector noise, the noise model is empirical rather than assumed; how well it represents non-stationary noise is limited by the amount of training noise (3040 s per stretch), and we test it here by repeating the analysis on 15 independent stretches rather than by assuming it. Third, the output is a full joint posterior $p(\Omega, \xi|x)$, enabling Bayesian model comparison and hierarchical inference.

A parallel line of recent work has examined the Gaussian likelihood assumption itself. Franciolini et al. [22] presented a systematic study of likelihood functions for SGWB searches, demonstrating that the Gaussian cross-correlation likelihood can introduce systematic biases when the number of data segments is small or the signal-to-noise ratio is moderate. Their finding that Gaussianity breaks down in precisely the regimes where our method excels—short segments, strong per-segment signals—provides independent theoretical motivation for likelihood-free approaches. Additionally, Karnesis et al. [23] proposed heavier-tailed likelihoods for characterizing non-Gaussian stochastic signals in the context of LISA, highlighting the broader community interest in moving beyond the Gaussian paradigm.

A particularly close methodological precedent is the SBI-for-SGWB framework of Alvey et al. [24], who applied truncated marginal neural ratio estimation (TM-

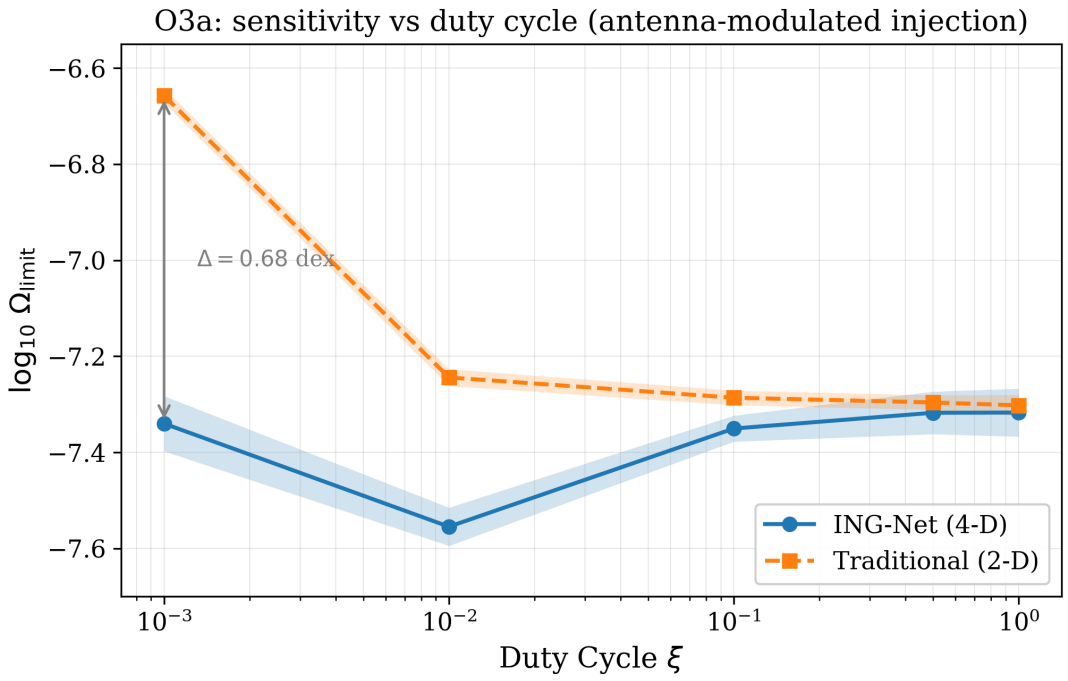


FIG. 7: [Figure to be regenerated from the revised analysis.] Sensitivity exclusion curves for the O3a dataset, showing the weakest detectable amplitude $\Omega_{\rm limit}$ as a function of duty cycle (lower is more sensitive). In the intermittent regime ($\xi < 10^{-2}$), ING-Net (blue) significantly outperforms the power-only baseline (orange), while both converge in the Gaussian limit ($\xi \to 1$). Shaded bands indicate $\pm 1\sigma$ over 5 independent training seeds. Results for O3b and O4a show the same pattern (Table II).

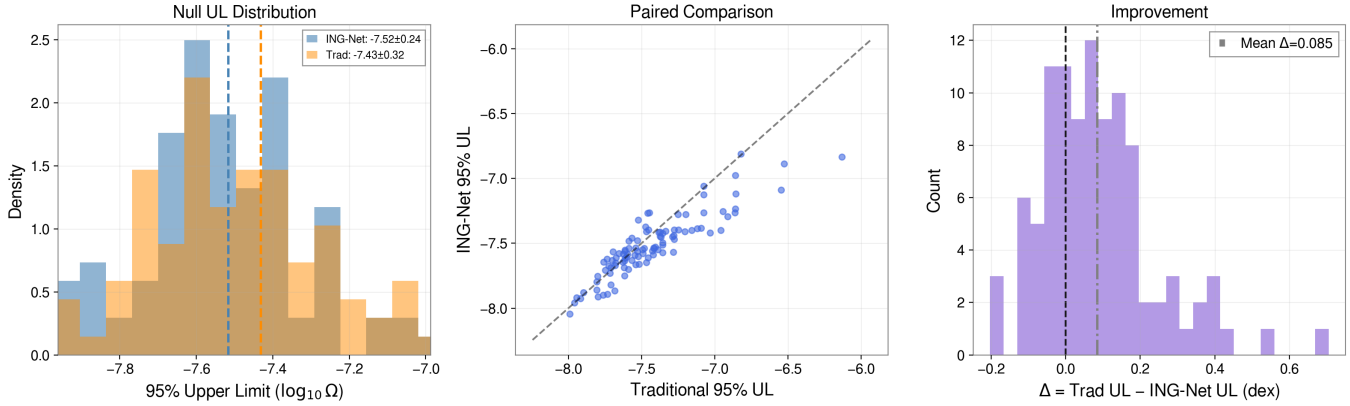


FIG. 8: Null result analysis for O4a. *Left:* Distribution of 95% upper limits on $\log_{10}\Omega$ from 100 noise-only segments for ING-Net (blue, mean -7.46) and the power-only baseline (orange, mean -7.37). *Center:* Paired per-segment comparison, showing that the two methods track each other closely on a segment-by-segment basis. *Right:* Distribution of the paired difference $\Delta_{\rm paired} = \log_{10}\Omega_{\rm UL}^{\rm Trad} - \log_{10}\Omega_{\rm UL}^{\rm ING\text{-}Net}$, centred at $+0.089$ dex, confirming that the inclusion of kurtosis does not introduce false signals on noise-only data. Results for O3a and O3b show the same qualitative behaviour with $|\Delta_{\rm paired}| \leq 0.05$ dex (Table III).

NRE) to mock LISA data and reproduced both template-based and agnostic Gaussian-SGWB searches. Our work is complementary in three respects: (i) we target the ground-based LIGO regime rather than LISA, so the relevant non-Gaussianity is driven by per-segment burst sparsity in $\sim$ 4 s windows rather than by the time-dependent instrument response; (ii) we use neural posterior estimation on a compact set of summary statistics rather than TMNRE on the full data spectrum, which makes the role of each summary explicit and enables the controlled ING-Net vs. Trad vs. Trad-FR comparison reported in Table II and Sec. VI A; and (iii) most

importantly, our feature vector includes the per-detector kurtosis (κ_{H1}, κ_{L1}), which directly probes the fourth cumulant and extends the SBI-for-SGWB programme into the *non-Gaussian / intermittent* regime that Alvey et al. do not address. The Alvey et al. framework and ours are therefore best viewed as covering complementary corners of the SGWB landscape—LISA / Gaussian / TMNRE on one side, LIGO / intermittent / NPE-with-kurtosis on the other.

Relation to the broader non-Gaussian SGWB landscape. Beyond the SBI-specific comparison above, our work sits within a larger body of literature on non-Gaussian and intermittent stochastic backgrounds, which we group into three lines. The first is the analytic / handcrafted-statistic line, beginning with Drasco and Flanagan [9], who first derived a maximum-likelihood detection statistic for popcorn-like intermittent backgrounds under a two-collocated-detector white-noise toy model; this was extended to realistic interferometer data by Thrane [21] through a generalised radiometer estimator, and to the fourth-cumulant / Edgeworth-expansion direction by Martellini and Regimbau [11], whose semi-parametric likelihood is the closest analytic relative of the kurtosis channel that ING-Net exploits. The second is the explicit-likelihood / importance-sampling line of Buscicchio et al. [25] and the improved-statistics follow-up by Ballelli et al. [26], who construct simulation-augmented detection statistics approaching the Neyman–Pearson optimum on toy non-Gaussian models. The third, most closely related in physical setting, is the Gaussian-mixture / stochastic-signal-based search of Lawrence et al. [27] for popcorn-like binary-black-hole backgrounds at LIGO/Virgo duty cycles $\xi \sim 10^{-3}$, and the deep-learning detector/regressor of Yamamoto, Kuroyanagi, and Liu [28], who applied CNN and residual-network classifiers to the Drasco–Flanagan toy model and directly estimated the duty cycle. From the population-physics side, Lamb and Taylor [29] have very recently derived analytic scaling relationships for the variance, skewness, and *kurtosis* of an SGWB from a compact-binary population, providing the astrophysical justification for treating higher cumulants as carriers of physical information rather than as nuisance properties of finite-sample estimators.

Our contribution is complementary to, rather than in competition with, all three lines. We do not derive a new model-specific maximum-likelihood statistic, importance-sampling statistic, or Gaussian-mixture estimator; instead we use simulation-based neural posterior estimation to infer the joint posterior $p(\Omega_{\rm GW}, \xi \mid \mathbf{x})$ from forward simulations. The key methodological distinction relative to all of the above is that we expose the fourth-cumulant information through *per-detector* excess-kurtosis summaries (κ_{H1}, κ_{L1}), which both carry the intermittency signature highlighted by Drasco–Flanagan and Yamamoto–Kuroyanagi–Liu and, by virtue of being per-detector rather than per-pair, provide an implicit coherency veto against single-detector glitches—a feature that, to our knowledge, none of the prior analytic, importance-sampling, Gaussian-mixture, or CNN-based intermittent-SGWB methods builds in by construction. This places ING-Net between the analytic non-Gaussian SGWB statistics and the recent SBI-based SGWB reconstruction methods, and extends the latter to the ground-based, low-duty-cycle, glitch-contaminated intermittent regime relevant for current LIGO data.

Within the rapidly growing SBI literature for gravitational-wave astronomy [12], our work occupies a distinct niche. Most existing SBI applications target parameter estimation of individual compact binary coalescence events [30, 31], where the signal model is well characterized and the primary challenge is computational speed. In contrast, our application targets a *stochastic* signal where the fundamental challenge is not speed but rather the extraction of information from higher-order statistics that are invisible to standard spectral methods.

Computational cost. [Timings and run counts to be updated for the revised analysis.] A separate question is how the cost of ING-Net compares to the cross-correlation pipeline it is meant to extend. The traditional Y-statistic for an SGWB search is dominated by per-segment FFTs of length matched to the analysis window, with a typical wall-clock cost on the order of CPU-minutes for a full O3a-scale dataset on a single workstation. ING-Net has a different cost profile: there is a one-time training cost—roughly 30 minutes of GPU time per configuration (one observing run, one feature set, one seed) on a single modern device, and ~ 9.7 GPU-hours total for the full multi-seed, multi-dataset, multi-feature-set sweep (5 seeds $\times$ 3 datasets $\times$ 2 configurations, distributed over 4 GPUs)—after which posterior evaluation for a single 4 s segment is a forward pass through the MAF and takes milliseconds. The training cost is therefore comparable to a single full run of the power-only pipeline on the same data and is amortised over all subsequent posterior queries, including the $N = 100$ noise-only realisations used for the null upper limits in Table III and the binary-search sweep used to compute the sensitivity limits in Table II. The simulator itself ($\sim 5 \times 10^4$ injected segments per training set) is also a one-time cost. For pipelines whose primary output is a single broadband upper limit on $\Omega_{\rm GW}$ from a fixed dataset, the power-only baseline remains the natural choice; the SBI cost only pays for itself when one wants the full (Ω, ξ) posterior, repeated sensitivity injections, or rapid evaluation on many segments, all of which are needed to make the kurtosis channel of Sec. V usable.

Frequency-resolved baseline: does higher dimensionality substitute for kurtosis?

A natural concern with the comparison in Table II is that the ING-Net advantage might reflect dimensionality rather than information content: the network sees a 4-D input while the power-only baseline sees only 2-D, and a richer input could in principle help even when the additional components carry redundant information. To address this directly we trained a third pipeline, denoted Trad-FR, in which the scalar cross-correlation C_{HL}

and log-power P are replaced by frequency-resolved analogues: C_{HL} becomes a per-bin coherence vector and P becomes a per-bin log-power vector, together producing an input of dimensionality comparable to or exceeding that of the ING-Net 4-D input. The training procedure, MAF architecture, and prior are otherwise identical. If the ING-Net advantage were merely a feature-count effect, Trad-FR should at least close part of the gap.

Because the per-bin coherence and log-power vectors contain the scalar summaries (C_{HL}, P) as simple functions of their components, an estimator of adequate capacity should do at least as well with the Trad-FR input as with the scalar one. This is what we find: on the three reference stretches, Trad-FR and the scalar Trad baseline reach the same $\Omega_{\rm limit}$ to within 0.1 dex at every duty cycle (Table IV), comparable to the uncertainty of a single-stretch limit. Frequency resolution applied to second-cumulant statistics therefore recovers the information on Ω that is already present in the scalar power summaries, but adds none on the duty cycle: the per-bin vectors remain second-cumulant quantities and are blind to the fourth cumulant, in which the intermittency signature resides. Dimensionality is thus not a substitute for cumulant order. One of the nine Trad-FR runs (O3a) diverged in training and is excluded. [Capacity scan (width and depth) for all three pipelines: pending.]

An emergent coherency veto from per-detector kurtosis. [Paragraph to be rewritten under referee comment 8.]

A feature of the per-detector summary statistics that we did not design for, but which emerges naturally from the choice to keep κ_{H1} and κ_{L1} as separate channels, is an implicit coherency veto against single-detector glitches. A genuine intermittent SGWB burst projects onto both detectors through their antenna patterns, so it raises the excess kurtosis in H1 and L1 together; an instrumental glitch is local to one site and raises the kurtosis of a single channel only. In the $(\kappa_{H1}, \kappa_{L1})$ plane these two cases occupy distinct regions—the signal along the diagonal $\kappa_{H1} \approx \kappa_{L1}$, a single-detector transient along one of the axes—whereas the cross-correlation C_{HL} alone does not separate them, since a glitch coinciding with cross-detector noise correlations can also elevate C_{HL}. Figure 9 illustrates this geometry. We stress that this is an interpretive observation about the information carried by the chosen features rather than an explicit processing step: the network receives only the four-dimensional summary and returns a posterior, and no glitch class is injected, labelled, or vetoed anywhere in the pipeline. The glitch locus in Fig. 9 is therefore drawn schematically; the empirical evidence that the kurtosis channel does not manufacture spurious signals on real glitch-contaminated data is the null-result analysis of Sec. V D, where the per-detector kurtosis measured on noise-only segments remains small ($|\hat{\kappa}_{H1,L1}| \lesssim 0.3$) and the null upper limits track the power-only baseline to within $|\Delta_{\rm paired}| \leq 0.09$ dex. A dedicated quantitative study of glitch rejection, using catalogued glitch samples and an explicit injection campaign, would make this veto property precise.

B. Signal model: assumptions, limitations, and robustness

Several simplifications in our signal model warrant discussion, as they define the regime of validity for our results.

Spectral shape. Our injected bursts are white (flat-spectrum) in the whitened data domain; the overall energy normalisation is calibrated against the theoretical strain spectrum of an isotropic background of fixed energy density, but no power-law spectral colouring is applied to the burst itself. Because the kurtosis statistic is a time-domain measure of how energy is distributed in time, it is insensitive to how the same energy is distributed in frequency, so the sensitivity limits in Table II are not tied to a particular spectral index. We quantify this explicitly in Sec. VI B, where colouring the bursts with a representative compact-binary index ($\alpha = 2/3$) shifts the recovered $\log_{10}\Omega$ by < 0.03 dex—well within the run-to-run statistical scatter of the sensitivity estimate. A fully waveform-consistent treatment, in which the injected events carry the spectral shape of a specific source class, is a natural refinement that we leave to future work.

Burst morphology and source parametrisation. Each event in our simulator is modelled as a windowed segment of white Gaussian noise, rather than as a physically motivated waveform (e.g., a chirp for BBH mergers or a cusp template for cosmic strings). Correspondingly, the simulator does not carry the internal parameters of a compact binary—component masses, spins, or luminosity distance—and there is therefore no compact-binary parameter distribution to specify: the only quantities sampled are the population-level parameters (Ω, ξ) and, per event, an isotropic sky direction and polarisation angle (see Sec. II). The injected signal is controlled at the level of its *energy density*: the per-event amplitude scale is set by Ω/ξ with the overall normalisation fixed by the energy calibration of Eq. (11), so it is the time-averaged energy density—not the power spectrum of a specific binary population—that is held consistent across the prior. This choice is deliberate: it ensures that the network's sensitivity to intermittency is driven by the *statistical* properties of the background (the duty cycle and amplitude distribution) rather than by the *deterministic* features of individual events. The cost is that our method does not exploit event-specific morphological information, which could provide additional discriminating power for specific source classes. A natural extension—which would also let the simulator answer population questions of direct astrophysical interest—is to replace the generic burst with physical compact-binary waveforms drawn from a mass/spin/redshift distribution, so that ξ and Ω become derived consequences of an explicit merger-rate model

TABLE IV: Frequency-resolved baseline comparison on the three reference stretches (antenna injection; seeds 1400–1402; [n/N] runs per column after excluding diverged runs). Values are $\log_{10}\Omega_{\rm limit}$. $\Delta_{\rm FR}$ is the offset of Trad-FR relative to the scalar Trad baseline; a positive entry indicates that the frequency-resolved features improve the limit.

ξ	ING-Net	Trad	Trad-FR	$\Delta_{\rm FR}$ (dex)
[revised values]				

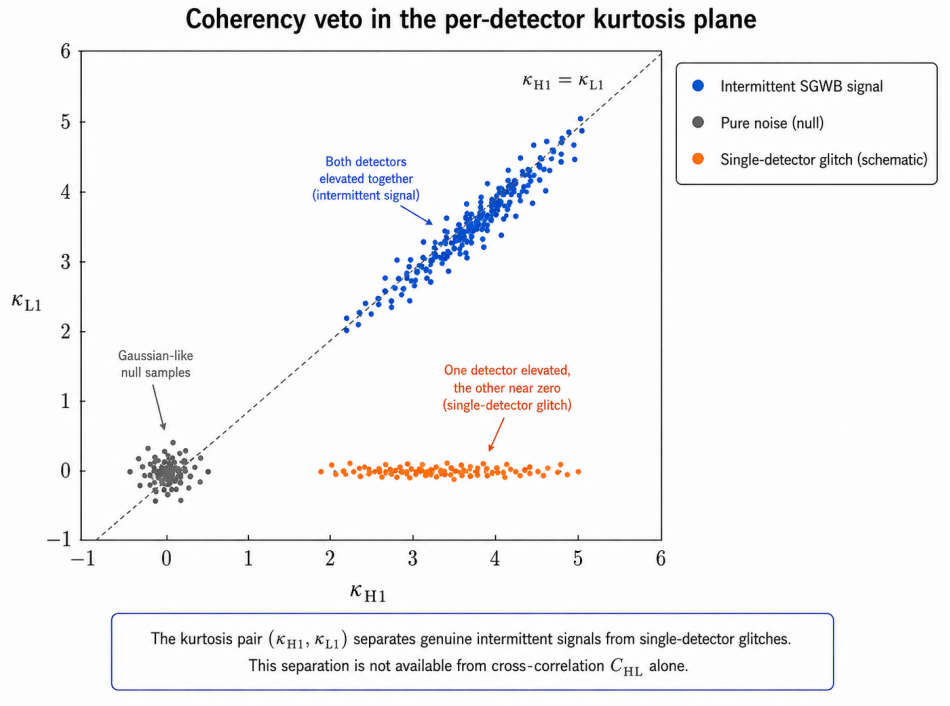


FIG. 9: Coherency veto in the per-detector kurtosis plane. Each point is a 4 s segment plotted by its measured excess kurtosis in the two detectors, $(\kappa_{H1}, \kappa_{L1})$. A genuine intermittent SGWB signal (blue) projects onto both detectors and populates the diagonal $\kappa_{H1} \approx \kappa_{L1}$ at elevated values; pure Gaussian-like noise (grey) clusters near the origin. A single-detector glitch elevates one channel only and therefore falls along an axis, far from the diagonal; this glitch locus is shown schematically (orange), as no glitch class is injected or labelled in our pipeline (see text). The cross-correlation C_{HL} alone cannot make this separation, whereas the per-detector kurtosis pair does—an emergent property of keeping the two kurtosis channels separate rather than a hand-designed veto.

rather than free population parameters; we leave this to future work.

We argue, however, that the generic-burst model is *sufficient* for the inference target of this work, and not merely a convenient approximation. The quantities our analysis conditions on are the low-order per-segment cumulants summarised by $(C_{HL}, P, \kappa_{H1}, \kappa_{L1})$: the second cumulant fixes the combination $\xi\Omega$ and the fourth cumulant carries the additional dependence on ξ that breaks the degeneracy (Sec. V E). For a fixed energy density and duty cycle, these cumulants depend on *how the energy is partitioned in time*—between sparse loud events and a quiet baseline—and not on the deterministic waveform of any individual event. Two intermittent backgrounds with the same (Ω, ξ) but different per-event morphologies therefore produce the same per-segment cumulants up to fourth order, and are indistinguishable to an estimator that conditions only on those cumulants. Replacing the white burst with a physical chirp would change the within-event time–frequency structure but not the segment-level cumulant statistics that drive the inference; the morphological information so added is precisely the information our summaries discard by construction. Two potential routes by which event morphology might nonetheless leak into the result—the spectral shape of the burst and the detector geometry—are exactly the two we have tested directly and found negligible (Sec. VI B and the antenna analysis below). Finally, our headline result is the *differential* sensitivity Δ between ING-Net and the power-only baseline, which are trained and evaluated on identical injections; any residual dependence on the signal model acts on both pipelines in the same direction and largely cancels in Δ. The simplified burst is thus a deliberate and defensible choice for isolating the intermit-

TABLE V: $\log_{10}\Omega_{\rm limit}$ under identical and antenna-pattern injection (median over seeds 1400–1402; explicit coincidence veto applied to both pipelines; Δ = Trad − ING-Net, median of the per-seed differences). One O3b Trad seed diverged under antenna injection and is excluded ($n = 2$).

		Identical			Antenna		
ξ	Run	ING-Net	Trad	Δ	ING-Net	Trad	Δ
10^{-3}	O3a	−4.96	−4.49	+0.47	−4.43	−4.03	+0.41
	O3b	−4.95	−4.57	+0.38	−4.55	−4.07	+0.48
	O4a	−4.98	−4.51	+0.47	−4.61	−4.06	+0.55
10^{-2}	O3a	−4.85	−4.69	+0.16	−4.80	−4.58	+0.22
	O3b	−4.87	−4.73	+0.14	−4.85	−4.60	+0.25
	O4a	−4.88	−4.69	+0.19	−4.90	−4.64	+0.26

tency signal, and the absolute limits we report should be read as limits under this well-defined population model rather than as forecasts for one specific binary population.

Effect of the antenna response. To attribute the effect of the antenna response, we repeated the analysis of the O3a, O3b and O4a reference stretches with an "identical" injection, in which the same signal realisation is added to both detectors, and report $\Omega_{\rm limit}$ for both pipelines under both schemes (Table V). In both schemes the mean per-detector signal power is the same by construction; the antenna response changes three things: the zero-lag H1–L1 correlation of the signal drops from 1 to $\gamma_{\rm HL}(0) \simeq -0.89$ and varies from event to event, which weakens C_{HL}; the per-event power fluctuates between detectors, which raises $\kappa_{\rm sig}$ at fixed mean power by $c_F \simeq 1.43$ [Eq. (5)]; and at $\xi = 10^{-3}$, where a segment typically contains a single event, that event is often projected weakly onto one of the two detectors. The net effect on the gain is small. At $\xi = 10^{-3}$ both pipelines become shallower by ≈ 0.4–0.5 dex and Δ changes by -0.06 to $+0.10$ dex; at $\xi = 10^{-2}$ ING-Net is essentially unaffected (≤ 0.05 dex) while the power-only baseline becomes shallower by 0.05–0.13 dex, so that Δ increases by ≈ 0.1 dex. The modest increase is thus attributable to the degradation of the C_{HL} channel in the baseline rather than to an enhancement of the kurtosis channel.

Remaining geometric simplifications. Two simplifications remain. We do not currently model the inter-detector light-travel-time delay (up to ~ 10 ms for H1–L1), and our sky-position draws are independent across segments rather than tied to a physical population. The first is justified at our segment lengths because the kurtosis estimator is essentially insensitive to a $\lesssim 10$ ms shift within a 4 s segment; the second is justified because we infer marginal posteriors over (Ω, ξ) for which sky position is a nuisance parameter and the antenna draws are already isotropic. Extending the simulator to a discrete population of catalogued sources, each with its own sky direction, would be straightforward and is a natural follow-up.

Segment length and stationarity. We analyze 4-second data segments, a choice that represents a trade-off between capturing sufficient burst statistics per segment and maintaining approximate noise stationarity. For very low duty cycles ($\xi = 0.001$), the expected number of bursts per 4-second segment is $\xi T/\tau \simeq 1.0$ [Eq. (3)], meaning the kurtosis estimate within a single segment has large variance. Our binary search procedure for the sensitivity limit (Section V) accounts for this by averaging over multiple independent trials and seeds. However, analyzing longer segments or stacking information across multiple segments could further improve sensitivity at the cost of increased sensitivity to noise non-stationarity. Recent work by Franciolini et al. [22] systematically analyzed how segment size affects duty-cycle estimation in the Gaussian-mixture likelihood framework; our 4-second choice falls within the regime they identify as approximately unbiased for $\xi \gtrsim 10^{-3}$.

Model misspecification. As with all SBI methods, our framework is susceptible to model misspecification: if the real data distribution deviates from the simulated training distribution, the posterior estimates may be biased [12]. We mitigate this risk by training on real detector noise (rather than simulated Gaussian noise), which ensures that the network encounters realistic instrumental artifacts during training. All evaluation is performed on noise that is time-disjoint from the training noise (Sec. II C), and the analysis is repeated on 15 independent stretches. Explicit misspecification tests across observing runs (e.g., training on O3a noise and evaluating on O4a noise) would further strengthen confidence in the method's robustness.

Robustness to spectral shape. [Test to be repeated with the revised pipeline, or the paragraph removed.] Because the injected bursts are white in the whitened domain (Sec. VI B), a natural question is whether the sensitivity limits would change if the bursts instead carried the power-law spectrum expected from a specific astrophysical source class. We address this directly with a controlled comparison. Using identical random seeds and identical whitened noise segments, we generate two sets of injected bursts that differ *only* in their spectral content: the white bursts used throughout this work, and bursts coloured in the whitened domain by a representative compact-binary index $\alpha = 2/3$ (applied as a frequency-domain filter $(f/f_{\rm ref})^{\alpha/2}$ with $f_{\rm ref} = 25$ Hz and total energy held fixed). For each of the three observing runs and three representative duty cycles $\xi \in \{10^{-3}, 10^{-2}, 10^{-1}\}$ we draw 5000 segments per configuration and compare the resulting per-detector summary statistics.

The two sets are statistically indistinguishable in the kurtosis channel: the largest change in the (transformed) per-detector kurtosis across all configurations is 0.044, and propagating the coloured features through the trained network shifts the recovered $\log_{10}\Omega$ by at most 0.027 dex. This is only about half the run-to-run statistical scatter of the sensitivity estimate (≈ 0.06 dex

over the averaging rounds), and remains negligible under a Bonferroni-corrected significance test across the eighteen configurations (maximum Cohen's $d \approx 0.04$). The reason is physical: the energy normalisation is held fixed between the two sets, so colouring only redistributes power across frequency, whereas the kurtosis measures how power is distributed in *time*; a time-domain fourth-cumulant statistic is therefore largely blind to the spectral shape. This is the quantitative basis for the claim, made throughout this work, that our sensitivity limits are not tied to a particular spectral index, and it indicates that a waveform-consistent treatment of the spectral shape—while a natural refinement—would not materially alter the limits reported here.

C. Boundary with single-event searches

[Section to be rewritten with the single-event SNR of Eq. (12); wording depends on the positioning under referee comment 3.]

The characteristic single-event amplitude $\rho_{\rm ev}$ at our detection limit is narrowly clustered around $\rho_{\rm ev} \approx 11.3$ (Sec. V C), a value that places our analysis at an interesting boundary with the existing menu of LIGO–Virgo–KAGRA short-duration searches. We briefly contextualise the three pipelines that are most directly adjacent to our regime, since each addresses a different facet of the same data. We stress at the outset that the network-coincidence SNRs quoted below for those pipelines are matched-filter or coherent excess-power statistics, not the whitened-domain amplitude $\rho_{\rm ev}$; the two are comparable in order of magnitude for a flat-spectrum burst but are not identical quantities.

Coherent WaveBurst (cWB) [32, 33] is an unmodelled, minimum-assumption pipeline that constructs coherent time–frequency excess-power triggers across the detector network. It is currently the most sensitive ground-based search for generic short transients of unknown waveform, and reports detections at network signal-to-noise ratios down to $\rho_{\rm net} \sim 8$–10 depending on the morphology class. Its outputs are individual event triggers ranked by a coherent likelihood statistic; it does not return a posterior on a population-level parameter such as a duty cycle.

Omicron + LIB (oLIB) [34] is similarly unmodelled but uses single-detector Q-transform triggers (Omicron) clustered into coincident candidates and ranked with a Bayesian sine-Gaussian likelihood. It is optimised for the burst regime SNR ~ 8 and above and shares the trigger-list output structure of cWB.

PyCBC [35, 36] is a template-matched-filter pipeline targeting compact binary coalescences. It is the most sensitive search for individual events that match a specific waveform family (BBH, BNS, NSBH); single-detector triggers above SNR ~ 5.5–8 are admissible, with the final detection statistic combining triggers across the network. By construction it cannot detect an event whose morphology lies outside its template bank.

A fraction of the loudest individual events that drive our $\xi = 10^{-3}$ detection in principle fall above the per-event detection thresholds of cWB and (for matching morphologies) PyCBC, so the question of how our analysis relates to these pipelines deserves explicit treatment. Three points distinguish our framework. *(i) Question answered.* Our output is a joint posterior $p(\xi, \Omega \mid \text{data})$ over the population-level parameters of an intermittent SGWB, with frequentist coverage validated on null data (Table III). Burst pipelines output ranked event triggers with associated false-alarm rates and do not, by themselves, deliver a posterior on (ξ, Ω). *(ii) Use of unresolved events.* Our 4-s segment-level summary statistics aggregate *all* events in the segment, resolved and unresolved alike. The unresolved tail—segments containing many sub-threshold events whose superposition raises the per-segment kurtosis without triggering any individual event search—contributes coherently to the inference and is, in the low-ξ regime, the dominant source of constraining power. By contrast, a pipeline that thresholds on individual events necessarily discards this tail. *(iii) Use of resolved events.* The handful of segments containing an above-threshold event contribute to our analysis via their elevated $(\kappa_{H1}, \kappa_{L1})$ values, but they are not separately fitted as a waveform; conversely, a burst pipeline that detects the same event extracts considerably more information from it (sky position, central frequency, possibly a waveform-class label) than our framework uses. This is what makes the two approaches complementary rather than competitive: each pipeline is optimal for the part of the event-population posterior it is designed to access.

The natural quantitative bridge between these regimes is a hierarchical analysis in which the burst-search trigger list contributes a likelihood factor on the resolved-burst rate, and our segment-level posterior on (ξ, Ω) contributes a likelihood factor on the full population. The two factors share parameters (a single ξ governing both the resolved and unresolved fractions of the population) and can in principle be combined into a unified posterior that exceeds the sensitivity of either component alone. We leave this hierarchical combination, which would also allow the burst-search false-alarm rate to be propagated coherently into the SGWB inference, to future work.

D. Implications for Astrophysical Sources

The improved sensitivity at low duty cycles is particularly relevant for two classes of sources.

Population III binary remnants. The first generation of stars (Population III) formed at redshifts $z \sim 10$–30 and are expected to have produced massive black holes with masses ~ 30–$300\,M_\odot$. The merger rate of their binary remnants is highly uncertain, with theoretical predictions spanning several orders of magnitude [4]. In the most conservative scenarios, the resulting SGWB would be deeply intermittent ($\xi \ll 0.01$), producing a

popcorn-like background that is invisible to standard cross-correlation searches. [Numbers to be updated.] The ~ 0.66 dex sensitivity improvement at $\xi = 10^{-3}$ that we demonstrate in Table II corresponds to a factor of ~ 4.6 improvement in the detectable energy density, directly extending the observational horizon for this primordial population and potentially enabling constraints on Population III star formation models that are currently inaccessible.

Cosmic string cusps and kinks. Individual gravitational-wave events from cosmic string loops are the prototypical intermittent source [10]. Current constraints on the string tension $G\mu$ are derived using either burst searches (targeting individual events) or Gaussian SGWB searches (for the confusion background) [37]. These two approaches probe complementary regimes: burst searches are sensitive when individual events are loud, and Gaussian searches are sensitive when the confusion limit is reached. Our method bridges this gap, targeting the intermediate regime of overlapping but statistically resolvable bursts. By capturing the non-Gaussian signature of partially overlapping bursts, our framework could tighten constraints on $G\mu$ in the regime between $\sim 10^{-12}$ and $\sim 10^{-15}$, where neither existing approach is optimal.

Prospects for third-generation detectors. This work has significant implications for third-generation detectors such as the Einstein Telescope (ET) and Cosmic Explorer (CE). In the 3G era, the astrophysical foreground from compact binary mergers is expected to be detectable as a confusion noise that obscures weaker cosmological backgrounds from inflation, phase transitions, and cosmic strings [38]. If this foreground is partially intermittent—as expected for high-redshift sources near the boundary of resolvability—our non-Gaussian framework could serve a dual purpose: (i) characterizing the foreground's statistical properties to enable accurate subtraction, and (ii) searching for residual non-Gaussian signatures of cosmological origin in the subtracted data.

The scalability of our approach to longer observation times and lower frequencies (relevant for space-based detectors such as LISA) is an open question. The 4-second segment length used here is well matched to the ∼Hz–kHz band of ground-based detectors, but millihertz sources in LISA may require segment lengths of hours to days, potentially straining the summary statistic approach. Developing hierarchical summary statistics that capture non-Gaussianity across multiple timescales is a natural extension of this work.

E. Methodological Extensions

The framework admits several natural extensions. The hand-crafted summary vector could be replaced by a learned embedding network—for example a convolutional network operating on the time–frequency representation—which would capture features beyond the fourth cumulant that our four-dimensional summary discards [12]. Promoting the spectral index α and reference frequency $f_{\rm ref}$ to inference parameters would let the spectral shape and the non-Gaussian properties be estimated jointly. And because the network returns a full posterior $p(\Omega, \xi \mid x)$, the non-Gaussian module can be folded into a hierarchical Bayesian analysis alongside a standard Gaussian cross-correlation search, with the latter supplying broadband amplitude constraints and the former breaking the rate–amplitude degeneracy.

VII. CONCLUSION

[Conclusion to be rewritten once the positioning under referee comment 3 is fixed.]

We have presented a framework for detecting and characterizing non-Gaussian, intermittent stochastic gravitational-wave backgrounds using Simulation-Based Inference. By moving beyond the Gaussian assumption inherent in standard cross-correlation searches, we have demonstrated that "intermittency is a resource" rather than a nuisance: the excess kurtosis produced by sparse, non-overlapping bursts carries extractable information about the source population that is invisible to spectral methods.

Our key results, validated on real LIGO data from three independent observing runs, are as follows. First, under the intermittent-signal model considered here, by conditioning the neural posterior estimator on individual-detector kurtosis statistics in addition to the cross-correlation and log-power, we alleviate the rate–amplitude degeneracy between signal rate ξ and amplitude Ω that limits all power-spectrum-based methods (Fig. 3b, Fig. 5). The power-only baseline, using the same network architecture but without kurtosis, fails to constrain the duty cycle ξ (Fig. 4). Second, in the intermittent regime ($\xi \leq 10^{-2}$), ING-Net achieves a sensitivity improvement of approximately 0.66 dex in $\Omega_{\rm GW}$ over the spectral baseline (Table II), with $\Delta = 0.683$ (O3a), 0.669 (O3b), and 0.640 (O4a) at $\xi = 10^{-3}$ when the H1–L1 antenna-pattern modulation is explicitly modelled. Each Δ is reported as the mean over 5 independent training seeds with per-cell $\sigma \leq 0.07$ dex, demonstrating robustness against GPU non-determinism in the NPE training. In the Gaussian limit ($\xi \to 1$) the two methods converge to within $|\Delta| \leq 0.039$ dex. Third, on noise-only data the non-Gaussian framework yields 95% upper limits on $\Omega_{\rm GW}$ that match the spectral baseline to within $|\Delta_{\rm paired}| \leq 0.09$ dex (Table III), demonstrating that the inclusion of higher-order statistics does not introduce false signals. Finally, the consistency of these results across O3a, O3b, and O4a—which span different detector configurations, noise floors, and calibration states—provides evidence that the method generalises beyond the specific training noise realisation.

We have also identified several limitations that de-

fine the current scope of validity. The injected bursts are white in the whitened domain and the signal model uses a simplified burst morphology rather than physical waveforms; we show in Sec. VI B that the spectral-shape choice has a negligible effect on the recovered limits, but a waveform-consistent treatment of specific source classes remains future work. We model H1–L1 antenna-pattern modulation directly in the simulator, which reproduces the standard overlap reduction function $\gamma_{\mathrm{HL}}(0) \approx -0.89$ but does not yet include the inter-detector light-travel-time delay or a discrete population of catalogued sources (Sec. VI B).

Extending the analysis to the full O4 dataset—of which the first part (O4a) has already yielded 128 compact-binary candidates ($p_{\mathrm{astro}} \geq 0.5$) in GWTC-4.0 [39]—and to the upcoming O5 run will sharpen these constraints. More broadly, the transition from Gaussian to non-Gaussian SGWB analysis represents a qualitative shift in the information we can extract from gravitational-wave data. As detector sensitivities improve and the accessible source population grows, methods that exploit the full statistical structure of the data—rather than its second moment alone—will become increasingly essential for disentangling the astrophysical and cosmological components of the gravitational-wave background.

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CODE AND DATA AVAILABILITY

The source code used to produce the results in this paper, including the simulator, training pipeline, and evaluation scripts, will be made publicly available at `https://github.com/YifangLuo/ing-net` upon acceptance. Strain data for the O3a, O3b, and O4a observing runs were obtained from the Gravitational Wave Open Science Center [1, 2, 39] at `https://gwosc.org`. The trained network checkpoints and the configuration files used to generate Tables II and III will be deposited in a Zenodo archive linked from the GitHub repository.

Appendix A: Selection of data stretches

Each observing run contributes one reference stretch (Table I) and four further stretches of 4096 s, drawn by the following rule, which was fixed before any of the additional stretches was analysed.

Candidate start times. With a fixed random seed (20261003), 300 integer GPS start times were drawn uniformly within each run: O3a 1238166018–1253977218, O3b 1256655618–1269363618, and O4a 1368975618–1389456018 (O4a counted from the start of the observing period, excluding the engineering run). The SHA-256 digest of the resulting candidate list is `a84de4b83e8d4af37ffe187e6d251e910447d9ab1d16af05376bf3517dba3a98`.

Eligibility. A candidate was accepted if (i) the GWOSC `DATA` flag covers the full 4096 s in both H1 and L1; (ii) no event published by GWOSC falls within the stretch, including a 2 s margin at either end; and (iii) the stretch is separated by at least 86 400 s from every other selected stretch of the same run, including the reference stretch. No data-quality category or glitch-related flag was used.

Selection. Candidates were examined in the order in which they were drawn, and the first four eligible ones were selected. No stretch was replaced after selection. A candidate would have been skipped only if the downloaded data were unusable (wrong length, NaN values, or identically zero).

The selected stretches are listed in Table VI.

TABLE VI: GPS start times of the 15 analysed stretches (4096 s each). The first entry of each run is the reference stretch.

	O3a	O3b	O4a
Reference	1238919012	1264612209	1369205931
2	[]	[]	[]
3	[]	[]	[]
4	[]	[]	[]
5	[]	[]	[]

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